RevA SCH V4 R11 / JST-EH THT modular I/O / long NET stubs / stock C2685112 / 0603 / 2-layer / NO PCB

GPIO15_UART_RX
R30
1k / LED current
RX_LED_ANODE
GND
D5
GREEN / RX activity
RX_LED_ANODE

MIC_ONBOARD_AC
C10
1uF / onboard MIC coupling
MIC_ONBOARD_OUT
MIC_ONBOARD_OUT
MK1
GND
3V3
Onboard analog MEMS MIC

C11
100nF / onboard MIC
GND
3V3

MIC_EXT_R_BIAS
C40
1uF / MIC R coupling
MIC_EXT_R_AC
J9
MIC R / A1251 1.25mm 2P
MIC_EXT_R_BIAS
R23
10k / MIC R bias
5V
GND

MIC_EXT_L_BIAS
C41
1uF / MIC L coupling
MIC_EXT_L_AC
J2
MIC L / A1251 1.25mm 2P
MIC_EXT_L_BIAS
R22
10k / MIC L bias
5V
GND

5V
C43
1uF / MIC supply
GND
5V
C42
1uF / MIC supply
GND

AMP_MODE
R5
5V
0R / MODE high

AMP_VREF
C3
1uF / VREF
GND

AMP_MUTE_N
R4
5V
0R / MUTE high

AMP_SHDN_N
R3
5V
0R / SHDN high

J11
AMP_L_P_RAW
GND
AMP_L_N_RAW
5V
AMP_MUTE_N
AMP_VDD_FLT
AMP_INL
AMP_VREF
+OUT_L
PGNDL
-OUT_L
PVDDL
MUTE#
VDD
INL
VREF
+OUT_R
PGNDR
-OUT_R
PVDDR
SHDN#
GND
INR
MODE
AMP_R_P_RAW
GND
AMP_R_N_RAW
5V
AMP_SHDN_N
GND
AMP_INR
AMP_MODE
PAM8406D

AMP_R_P_RAW
FB3
120R@100MHz 2A
SPK_R_P
SPK_R_P
C9
220pF
GND

AMP_R_N_RAW
FB2
120R@100MHz 2A
SPK_R_N
SPK_R_N
C8
220pF
GND

J1
SPK R / A1251 1.25mm 2P
SPK_R_N
SPK_R_P

AMP_INR_AC
C2
1uF / right AC coupling
CODEC_RIGHT_LOP
AMP_INR_AC
R2
10k / INR
AMP_INR

AMP_INL_AC
C1
1uF / left AC coupling
CODEC_LEFT_LOP
AMP_INL_AC
R1
10k / INL
AMP_INL

J16
GND
BATT_IN
DXL_DATA
DXL / A2007 2.0mm 3P

J17
GND
BATT_IN
DXL_DATA
DXL / A2007 2.0mm 3P

C17
100nF / U6
GND
3V3

C16
100nF / U5
GND
3V3

C18
100nF / U7
GND
3V3

DXL_DATA_CLAMPED
D4
GND
5V1
DXL_DATA_INT
R33
0Ω
DXL_DATA_CLAMPED
DXL_DATA_CLAMPED
TH1
0Ω
DXL_DATA

3V3
R29
10k / RX gate enable
RX_GATE_EN

GND
R28
20k / direction pull-down
DXL_DIR

U6
DXL_DIR
DXL_DATA_INT
GND
3V3
TTL_RX_RAW
SN74LVC1G125DBVR

U5
TTL_RX_RAW
RX_GATE_EN
GND
3V3
GPIO15_UART_RX
74LVC1G08 / RX gate

GPIO14_UART_TX
R27
10k / direction base
DXL_DIR_B

U7
DXL_DIR
GPIO14_UART_TX
GND
3V3
SN74LVC1G126 / TX

DXL_DIR_B
Q1
MMBT3906 / direction
DXL_DIR
DXL_DATA_INT
3V3

3V3
R31
10k / RX pull-up
TTL_RX_RAW

GPIO14_UART_TX
R26
10k / UART TX pull-up
3V3

3V3
R32
10k / data pull-up
DXL_DATA_INT

DYNAMIXEL TTL HALF-DUPLEX — RS-485 REMOVED

CLASS-D SPEAKER + MICROPHONES

AMP_VDD_FILT
C34
10nF
GND

5V
C5
10uF
GND

5V
C7
10uF
GND

AMP_VDD_FILT
FB1
600R@100MHz 1.5A
5V
AMP_VDD_FILT
C33
1uF
GND

5V
C4
1uF
GND

5V
C6
1uF
GND

GND
C28
100nF
3V3

GND
C30
100nF
3V3

GND
C32
100nF
3V3

GND
C36
100nF
3V3

3V3
C27
1uF
GND

3V3
C29
1uF
GND

3V3
C31
1uF
GND

3V3
C35
1uF
GND

1V8
C12
10uF / 1V8
GND
C26
100nF / 1V8
1V8

3V3
C13
10uF / 3V3
GND

3V3
C37
10uF / 3V3
GND

U3
GND
1V8
VIN
3V3
XC6206P182 / 1V8

U2
GPIO19_PCM_FS
GPIO18_PCM_CLK
GPIO21_PCM_DOUT
GPIO20_PCM_DIN
AUDIO_MCLK
GPIO03_I2C_SCL
GPIO02_I2C_SDA
CODEC_RESET_N
WCLK
BCLK
DIN
DOUT
MCLK
SCL
SDA
RESET#
HPLOUT
HPROUT
HPLCOM
HPRCOM
AVDD
DRVDD
DRVDD
IOVDD
DVDD
MIC1LP/LINE1LP
MIC1LM/LINE1LM
MIC1RP/LINE1RP
MIC1RM/LINE1RM
LEFT_LOP
LEFT_LOM
RIGHT_LOP
RIGHT_LOM
MIC2R/LINE2R
MIC2L/LINE2L/MICDET
MICBIAS
EP
AVSS1
AVSS2
DRVSS
DVSS
MIC_EXT_L_AC
GND
MIC_EXT_R_AC
GND
CODEC_LEFT_LOP
CODEC_RIGHT_LOP
MIC_ONBOARD_AC
MIC2L_GND_LINK
MICBIAS
TLV320AIC3104IRHBR

GND
R7
0R / MIC2L to GND
MIC2L_GND_LINK

GND
C58
100nF / codec reset
CODEC_RESET_N

GND
C14
100nF / oscillator
3V3

3V3
Y1
Tri-state
GND
VDD
OUT
3V3
AUDIO_MCLK
12MHz

3V3
R6
10k / codec reset
CODEC_RESET_N

AUDIO CODEC + 1V8

J4
5V
5V
GND
GPIO14_UART_TX
GPIO15_UART_RX
GPIO18_PCM_CLK
GPIO20_PCM_DIN
GPIO21_PCM_DOUT
3V3
GPIO02_I2C_SDA
GPIO03_I2C_SCL
GND
GPIO19_PCM_FS
GND
FH-00339 stock pins / Zero pair-map / bottom

3V3
R13
10k / SCL pull-up
GPIO03_I2C_SCL

3V3
R12
10k / SDA pull-up
GPIO02_I2C_SDA

J5
GND
3V3
GPIO02_I2C_SDA
GPIO03_I2C_SCL
GND
ToF / Qwiic / pad 0 GND

D1
GND
5V_PRE_IDEAL
BAT545W / diode 1 used

5V_PRE_IDEAL
C22
22uF 10V X5R
GND

BATT_IN
C15
10uF
GND

5V_PRE_IDEAL
C21
22uF 10V X5R
GND

IDEAL_VS
C39
100nF 50V
GND

U10
IDEAL_VS
GND
5V
IDEAL_GATE
5V_PRE_IDEAL
LM5050-1

BUCK_BST
C20
100nF 50V / bootstrap
BUCK_SW

5V
R8
100R
IDEAL_VS

U9
5V_PRE_IDEAL
BATT_IN
BATT_IN
BUCK_BST
BUCK_SW
5V_PRE_IDEAL
GND
AP63205 / 5V buck

L1
BUCK_SW
6.8uH / 3A / Isat 4.3A
IDEAL_GATE

5V
Q2
SI2312 / ideal diode MOS
5V_PRE_IDEAL

ZERO EVEN PIN n -> J4 PAD n-1
ZERO ODD PIN n -> J4 PAD n+1
J4 = STOCK FH-00339 / C2685112 PAD NUMBERS
40-PIN HAT + J5 TOF / QWIIC

POWER — OFFICIAL AP63205 + LM5050-1

MICRODUCK HAT REVA — OFFICIAL C1 DERIVATIVE

Sheet: /
File: ProPrj_HAT_REVA_SCH_2026-10-15.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. 10.99.0-6189-g113c71d59d
Id: 1/1

嘉立创EDA